C.1.3. JOINT PROCESSES COMBINE THESE SUBSYSTEMS

Examples in the main text used three Mess3 factors and two Bloch Walk factors. The join representation spans $3^5 = 243$ dimensions, (but varies in only 242 dimensions, since it is restricted to the 242-simplex of this 243-dimensional space). The factored representation spans $5 \times 2 = 10$ dimensions, and can be seen as the Cartesian product of three 2-simplices Δ^2 and two 2-balls $\circ^2$, with a collective space $\Delta^2 \times \Delta^2 \times \Delta^2 \times \circ^2 \times \circ^2$.

C.2. Independent

Here we describe the independent factors experiments that test whether transformers learn factored representations when trained on sequences generated by a product of independent stochastic processes. In this setting, we compose five factors—three Mess3 processes operating on 2-simplices and two Bloch Walk processes operating on Bloch spheres—where each factor evolves according to its own transition dynamics and emits tokens independently.

Each factor n has a 3-dimensional latent space and emits sub-tokens $z^{(n)} \in \mathcal{Z}_n$ according to its own emission distribution. For the Mess3 factors ($n \in \{1, 2, 3\}$), the sub-tokens are $\mathcal{Z}_n = \{0, 1, 2\}$; for the Bloch Walk factors ($n \in \{4, 5\}$) the sub-tokens are, $\mathcal{Z}_n = \{0, 1, 2, 3\}$. Sub-tokens combine into a single observed token by mapping the Cartesian product to integers: $x \in \mathcal{X} = \{0, 1, \ldots, 431\}$, giving a vocabulary of size $|\mathcal{X}| = 3^3 \times 4^2 = 432$. As an explicit construction of the tokens from sub-tokens, we can take $x_\ell = \left(\sum_{n=1}^{3} z_\ell^{(n)} 3^{n-1}\right) + \left(\sum_{n=4}^{5} z_\ell^{(n)} 3^3 4^{n-4}\right) \in \mathcal{X} = \{0, 1, \ldots, 431\}$.

Because the factors are independent, the joint transition operator takes the tensor-product form

$$T^{(x)} = \bigotimes_{n=1}^{N} T_n^{(z^{(n)})}, \tag{31}$$

and predictive vectors remain product states throughout belief updating. The joint latent space is the tensor product of the five 3-dimensional factor spaces, giving dimension $3^5 = 243$. Since predictive vectors are normalized, they vary in $243 - 1 = 242$ dimensions. In contrast, the factored representation tracks each factor's predictive vector separately in orthogonal subspaces; since each factor's predictive vector is also normalized, each varies in $3 - 1 = 2$ dimensions, giving $5 \times 2 = 10$ total dimensions.

Table 1. Process configurations for the independent factor experiments.

Factors	**Process**	**Parameters**
F_0, F_1, F_2	Mess3	$\alpha = 0.6,\ x = 0.15$
F_3, F_4	Bloch Walk	$\alpha = 1.0,\ \beta = 3.0$

Table 1 summarizes the process parameters. Please notice and forgive the abuse of notation for both x and α.

C.3. Conditionally Independent

Because the condition for a lossless projection to the product state is *conditional* independence, we expect that models will find the factored representation when the factors depend on each other, provided their joint evolution can be described by equation (4). To investigate this we investigate a feed-forward dependency chain where each factor's dynamics in the latent space depend on preceding factors' outputs, but tensor-product structure is preserved: $T^{(x)} = \bigotimes_{n=1}^{N} T_n^{(z^{(n)}|z^{(1:n-1)})}$.

Following section A.4, there must exist a lossless linear map from the product-state sub-manifold to a factored direct-sum representation where the local predictive vectors for each subspace live in orthogonal subspaces:

$$\boldsymbol{\eta}_{\text{FWH}}^{(x_{1:\ell})} = \bigoplus_{n=1}^{N} \boldsymbol{\eta}_n^{(x_{1:\ell})} = (\tilde{\boldsymbol{\eta}}_1^{(x_{1:\ell})}, \tilde{\boldsymbol{\eta}}_2^{(x_{1:\ell})}, \ldots, \tilde{\boldsymbol{\eta}}_N^{(x_{1:\ell})}) \,. \tag{32}$$

We train a transformer on data generated from a chain of three Mess3 processes and two Bloch Walk processes, similar to our earlier experiments—but now each of the subprocesses depends on [all of] the factors above it in the chain. As training data, we use tokens x with a one-to-one correspondence to the Cartesian product of sub-tokens.

Table 2. Process configurations for the conditionally independent factor experiments. Left: parameters for each process variant. Right: variant selection for each factor, where the control variable C is the sub-token emitted by the preceding factor.

ID	Process	Parameters
M1	Mess3	$x = 0.15,\ \alpha = 0.60$
M2	Mess3	$x = 0.11,\ \alpha = 0.79$
M3	Mess3	$x = 0.50,\ \alpha = 0.60$
B1	Bloch Walk	$\alpha = 1.0,\ \beta = 2.00$
B2	Bloch Walk	$\alpha = 1.0,\ \beta = 2.50$
B3	Bloch Walk	$\alpha = 1.0,\ \beta = 3.00$
B4	Bloch Walk	$\alpha = 1.0,\ \beta = 3.50$

Factor	Control	Variants
F_0	—	M1
F_1	$C \in \{0, 1, 2\}$	M1, M2, M3
F_2	$C \in \{0, 1, 2\}$	M1, M2, M3
F_3	$C \in \{0, 1, 2\}$	B1, B2, B3
F_4	$C \in \{0, 1, 2, 3\}$	B1, B2, B3, B4

Table 2 shows the parameters for each process variant and the mapping from control sub-tokens to variants. The variation in β for Bloch Walk and in both x and α for Mess3 ensures that different variants induce meaningfully different dynamics, testing whether the model can track factor-specific predictive vectors that evolve under context-dependent transition rules.

C.4. Almost-independent example: Noised factors

Here we describe the noised factors experiments that test whether transformers maintain factored representations even when the data is not fully consistent with a factorized form. In this setting, the optimal predictive distribution lies off the product-state manifold in the joint belief space, meaning that cross-entropy loss can in principle be decreased by leveraging the higher-dimensional joint space rather than maintaining strict factorization.

This is an example of an indecomposable process, which has the general form: $T^{(x)} = \varepsilon T_{\text{int}} + (1-\varepsilon) \bigotimes_{n=1}^{N} T_n^{(x)}$.

We use the same five independent factors as described above, but now each emitted token $x \in \mathcal{X} = \{0, 1, \ldots, 431\}$ passes through a memoryless noisy channel with probability ε of being replaced uniformly by any other token.

To describe the token-labeled transition matrices explicitly, it is useful to define $\mathcal{Z} = \bigtimes_{n=1}^{N} \mathcal{Z}_n$. The transition operators on the joint latent space then take the form

$$T^{(x)} = (1-\varepsilon)\Big(\bigotimes_{n=1}^{N} T_n^{(z^{(n)})}\Big) + \frac{\varepsilon}{|\mathcal{X}|-1} \sum_{z'^{(1:N)} \in \mathcal{Z} \setminus \{z^{(1:N)}\}} \bigotimes_{n=1}^{N} T_n^{(z'^{(n)})} \,. \tag{33}$$

We train models across noise levels $\varepsilon \in \{0, 0.001, 0.01, 0.05, 0.1, 0.2, 0.3, 0.5\}$, spanning from the fully factored regime ($\varepsilon = 0$) to a highly corrupted setting where half of all tokens are randomized.

For low ε, we find that the model first learns a factored representation, which does a relatively good job at prediction with relatively few dimensions. To get below a critical loss threshold (which is a function of ε), the model then slowly increases the number of effective dimensions of its residual stream that it uses.

What is going on? For low ε, this noised process is *almost* product preserving, and so the predictive vectors remain close to the product-state manifold. The projection down to the factored representation is not too lossy. But, as the model receives pressure to lower loss, or if ε is large enough such that the projection is quite lossy, the model is pressured to learn the higher-dimensional representation that can losslessly encode off-manifold predictive vectors.

D. Extended Experimental Results

D.1. Details of Independent Experiments

In the main text we show that, over training, the model learns a structure with dimensionality and geometry that is consistent with the factored representation. In Figure 8, we show additionally that the formation of this structure during training is associated with a reduction in loss. See the figure caption for additional details.

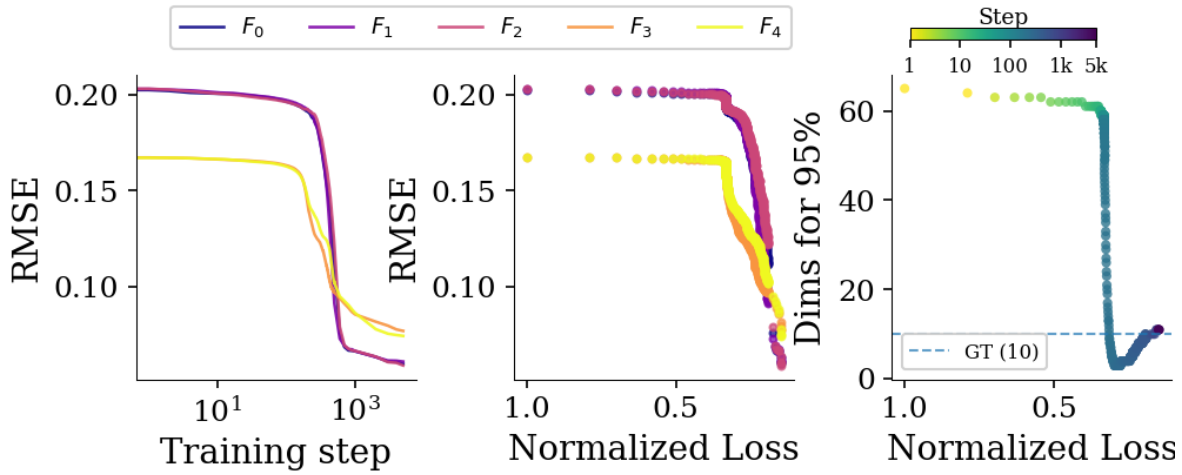

Figure 8. Early training dynamics. RMSE to the 5 independent factors, the number of dimensions needed to reach 95% CEV, and loss all decrease monotonically over training. (left) RMSE to each factor decreases monotonically over training (center) **RMSE to the predicted ground truth geometry is correlated with loss: as the regression improves, the loss consistently decreases.** (right) The model is able to consistently reduce loss by dropping dimensions until about 1000 steps. After this, it continues to reduce loss by converging to a dimensionality consistent with the factored representation. Further training results in a long slow decrease of loss and RMSE towards 0 with minimal change in effective dimensionality.

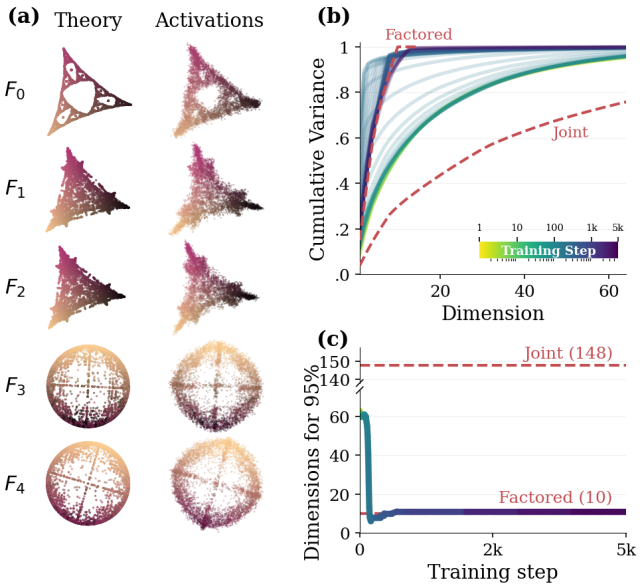

Figure 9. **Transformers learn to factor the world into parts, even when the parts form a chain of dependencies.** We train a transformer on sequences generated by five conditionally-independent factors; the model sees only tokens corresponding to the Cartesian product of factor outputs. (a) Left: ground-truth belief geometry for each factor. Right: linear regression from activations recovers each geometry with high fidelity. (b) Cumulative explained variance (CEV) over training. Dashed lines show predictions for factored (10D) versus joint (242D) representations; the model converges to the factored prediction. (c) Effective dimensionality (dimensions for 95% variance) collapses to 10 during training, far below the joint requirement.

D.2. Conditionally-independent factors

Figure 9 reveals that that neural networks trained via next-token prediction on the resulting sequences automatically learn to factor their representation into parts, where uncertainty over the latent states of each part is encoded geometrically.